Homework 3

1. Determine the convergence and absolute convergence of the following series:

(1) $\sum_{n=2}^{\infty} \frac{i^n}{\ln n}$; (2) $\sum_{n=1}^{\infty} \frac{i^n}{n}$.

2. Try to identify the convergence regions for the following series:

(1) $\sum_{n=1}^{\infty} z^{n!}$; (2) $\sum_{n=1}^{\infty} \left(\frac{z}{1+z}\right)^n$;

(3) $\sum_{n=1}^{\infty} (-)^n (z^2+2z+2)^n$; (4) $\sum_{n=1}^{\infty} 2^n \sin\frac{z}{3^n}$.

3. Try to find the convergence radius of the following military ranks:

(1) $\sum_{n=1}^{\infty} \frac{1}{n^n} z^n$; (2) $\sum_{n=1}^{\infty} \frac{1}{2^n n^n} z^n$;

(3) $\sum_{n=1}^{\infty} \frac{n!}{n^n} z^n$; (4) $\sum_{n=1}^{\infty} \frac{(-)^n}{2^{2n}(n!)^2} z^n$;

(5) $\sum_{n=1}^{\infty} n^{\ln n} z^n$; (6) $\sum_{n=1}^{\infty} \frac{1}{2^{2n}} z^{2n}$;

(7) $\sum_{n=1}^{\infty} \frac{\ln n^n}{n!} z^n$; (8) $\sum_{n=1}^{\infty} \left(1-\frac{1}{n}\right)^n z^n$.

4. Expand the following function into a Taylor series at the specified point and give its radius of convergence:

(1) $1-z^2$, expand at $z=1$;

(2) $sinz$, expand at $z=n\pi$;

(3)$\frac{1}{1+z+z^2}$, expand at $z=1$;

(4) $\frac{sinz}{1-z}$, expand at $z=0$;

(5) exp $\{\frac{1}{1-z}\}$, expand at $z=0$;(You can only ask for the first four)

5. Expand the following function to a Taylor series at the specified point and give its convergence radius:

(1) lnz, expand at $z=i$, when $0 \ll \arg z < 2\pi$;

(2) lnz, expand at $z=i$, when $ln\, z|_{z=i} =-\frac{3}{2}\pi i$;

(3) The principal value of $arctanz$, expanded at $z=0$;

(4) $ln\frac{1+z}{1-z}$, expand at $z=\infty$, when $ln\frac{1+z}{1-z}\Big|_{z=\infty} = (2k+1)\pi i$;

6. Find the sum of the following infinite series, taking care to give the corresponding convergence region:

(1) $\sum_{n=0}^{\infty} \frac{1}{2n+1} z^{2n+1}$;

(2) $\sum_{n=0}^{\infty} \frac{1}{(2n)!} z^{2n}$;

(3) $\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(n+m)!}{n!\, m!} \left(\frac{z}{2}\right)^{n+m}$;

(4) $\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{(n+m+p)!}{n!\, m!\, p!} \left(\frac{z}{3}\right)^{n+m+p}$.

7. Find the Laurent expansion of the following function:

(1) $\frac{1}{z^2(z-1)}$, expand around $z = 1$;

(2) $\frac{1}{z^2(z-1)}$, the expansion area is $1 < |z| < \infty$;

(3) $\frac{1}{z^2-3z+2}$, the expansion area is $1 < |z| < 2$;

(4) $\frac{1}{z^2-3z+2}$, the expansion area is $2 < |z| < \infty$;

(5) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, the expansion area is $3 < |z| < 4$;

(6) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, the expansion area is $4 < |z| < \infty$;

8. Determine the nature of the isolated singularity of the following functions, and in the case of the poles, determine its order:

(1) $\frac{1}{z^2+a^2}$, $a \neq 0$;

(2) $\frac{\cos az}{z^2}$;

(3) $\frac{\cos az - \cos bz}{z^2}$, $a^2 \neq b^2$;

(4) $\frac{\sin z}{z^2} - \frac{1}{z}$;

(5) $\cos \frac{1}{\sqrt{z}}$;

(6) $\frac{\sqrt{z}}{\sin \sqrt{z}}$;

(7) $\frac{1}{(z-1)\ln z}$;

(8) $\int_0^z \frac{\sinh \sqrt{\zeta}}{\sqrt{\zeta}} \mathrm{d}\zeta$.